• Serii de puteri. Funcții analitice.

1. Să se dezvolte în serie de puteri în jurul punctului $z_0 = 0$, determinând și domeniul de convergență:

(a) $\frac{1}{2z-5}$ (d) $\frac{1}{(z^2-1)(z^2-9)}$ (g) $\sin^2 z$

(b) $\frac{1}{1+z^4}$ (e) $\frac{1+iz}{1-iz}$

(c) $\frac{1}{1-z+z^2}$ (f) $\frac{1}{(1+z)^2}$

Rezolvare

(a) $\frac{1}{2z-5} = \frac{1}{5}\cdot\frac{1}{\frac{2}{5}z-1} = \frac{-1}{5}\cdot\frac{1}{1-\frac{2}{5}z} = -\frac{1}{5}\sum_{n\geq 0}\left(\frac{2}{5}\right)^n\cdot z^n$; $\left|\frac{2}{5}z\right|<1$

$\Rightarrow -1 < \frac{2}{5}z < 1 \Rightarrow -\frac{5}{2} < z < \frac{5}{2} \Rightarrow |z| < \frac{5}{2}$

(b) $\frac{1}{1+z^4} = (1+z^4)^{-1} = 1 + \sum_{n\geq 1}\frac{(-1)(-2)(-3)\dots(-1-(n-1))}{n!}\cdot z^{4n}$; $|z|<1$

$= 1 + \sum_{n\geq 1}\frac{(-1)^n\cdot n!}{n!}\cdot z^{4n} = 1 + \sum_{n\geq 1}(-1)^n\cdot z^{4n} = \sum_{n\geq 0}(-1)^n\cdot z^{4n}$

(c) $\frac{1}{1-z+z^2} = \frac{1+z}{1+z^3} = (1+z)\sum_{n\geq 0}(-1)^n z^{3n}$; $|z|<1$

$= (1+z)(1-z^3+z^6-z^9+\dots) = (1+z-z^3-z^4+z^6+z^7-\dots) =$

$= \sum_{n\geq 0}(-1)^n\left(z^{3n}+z^{3n+1}\right)$; $|z|<1$

(d) $\frac{1}{(z^2-1)(z^2-9)} = \frac{1}{8}\cdot\frac{z^2-1-(z^2-9)}{(z^2-1)(z^2-9)} = \frac{1}{8}\left(\frac{1}{z^2-9}-\frac{1}{z^2-1}\right) =$

$= \frac{1}{8}\left[\frac{1}{9}\cdot\frac{1}{\frac{z^2}{9}-1} - \frac{1}{z^2-1}\right] = \frac{1}{8}\left[-\frac{1}{9}\cdot\frac{1}{1-z^2/9} + \frac{1}{1-z^2}\right] =$

$= \frac{1}{8}\left(\sum_{n\geq 0} z^{2n} - \frac{1}{9}\sum_{n\geq 0}\left(\frac{1}{9}\right)^n\cdot z^{2n}\right) =$

$= \frac{1}{8}\left(\sum_{n\geq 0} z^{2n} - \frac{1}{9}\sum_{n\geq 0}\left(\frac{z}{3}\right)^{2n}\right) =$

$= \frac{1}{8}\sum_{n\geq 0} z^{2n}\left(1-\frac{1}{9\cdot 3^{2n}}\right) = \frac{1}{8}\sum_{n\geq 0} z^{2n}\left(1-\frac{1}{9^{n+1}}\right)$; $|z|<3$

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(e) $\frac{1+iz}{1-iz} = (1+iz)\cdot\frac{1}{1-iz} = (1+iz)\sum_{n\geq 0} i^n z^n$; $|z|<1$

$= \sum_{n\geq 0} i^n z^n + \sum_{n\geq 0} i^{n+1} z^{n+1} = 1 + 2\sum_{n\geq 0} i^n z^n$

(f) $\frac{1}{(1+z)^2} = (1+z)^{-2} = 1 + \sum_{n\geq 1} \frac{(-2)(-3)(-4)\dots(-2-(n-1))}{n!} z^n$; $|z|<1$

$= 1 + \sum_{n\geq 1} \frac{(-1)^n (n+1)!}{n!} \cdot z^n = \sum_{n\geq 0} (-1)^n (n+1)(z^n)$.

g) $\sin^2 t = \frac{1-\cos 2z}{2} = \frac{1}{2} - \frac{1}{2}\sum_{n\geq 0} (-1)^n \cdot \frac{2^{2n}\cdot z^{2n}}{(2n)!}$; $z\in\mathbb{C}$

$= \sum_{n\geq 1} (-1)^{n+1} \cdot \frac{2^{2n-1}\cdot z^{2n}}{(2n)!}$

2. Să se dezvolte în serie de puteri următoarele funcții în jurul punctelor indicate:

(a) $\frac{1}{1-z}$; $z_0 = -1$ (b) e^{2z} ; $z_0 = 1$

Rezolvare

(a) $\frac{1}{1-z} = \frac{1}{1-z+1-1} = \frac{1}{2-(z+1)} = \frac{1}{2}\cdot\frac{1}{1-\frac{z+1}{2}} =$

$= \frac{1}{2}\sum_{n\geq 0} \frac{(z+1)^n}{2^n} = \sum_{n\geq 0} \frac{(z+1)^n}{2^{n+1}}$; $|z+1|<2$

(b) $e^{2z} = e^{2(z-1)+2} = e^2 \cdot e^{2(z+1)} = e^2 \cdot \sum_{n\geq 0} \frac{(2z+2)^n}{n!} =$

$= e^2 \sum_{n\geq 0} \frac{(z-1)^n \cdot 2^n}{n!}$; $z\in\mathbb{C}$

- Serii Laurent.

1. Să se dezvolte în serie Laurent pe domeniile indicate funcțiile:

(a) $\frac{1}{(z^2-1)^2}$ pe $0<|z-1|<2$ (b) $\frac{1}{z(z-1)^2}$ pe $0<|z-1|<1$.

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Rezolvare

(a) $\frac{1}{(z^2-1)^2} = \frac{1}{(z-1)^2}\cdot\frac{1}{(z+1)^2} = \frac{1}{(z-1)^2}\cdot\frac{1}{[(z-1)+2]^2} =$

$= \frac{1}{(z-1)^2}\cdot\frac{1}{4\left[\frac{z-1}{2}+1\right]^2} = \frac{1}{(z-1)^2}\cdot\frac{1}{4}\cdot\left(1+\frac{z-1}{2}\right)^{-2} =$

$= \frac{1}{4(z-1)^2}\cdot\sum_{n\geq 0}\frac{(-2)(-3)(-4)\dots(-2-(n-1))}{n!}\cdot\left(+\frac{1}{2}\right)^n(z-1)^n =$

$= \frac{1}{4(z-1)^2}\cdot\sum_{n\geq 0}\frac{(-1)^n(n+1)!}{n!}\left(+\frac{1}{2}\right)^n(z-1)^n =$

$= \frac{1}{4(z-1)^2}\cdot\sum_{n\geq 0}\left(-\frac{1}{2}\right)^n(n+1)(z-1)^n =$

$= \frac{1}{4}\left(\sum_{n\geq 0}\left(-\frac{1}{2}\right)^n(n+1)(z-1)^{n-2}\right.$

(b) $\frac{1}{z(z-1)^2} = \frac{1}{(z-1)^2}\cdot\frac{1}{(z-1)+1} = \frac{1}{(z-1)^2}\cdot[1+(z-1)]^{-1} =$

$= \frac{1}{(z-1)^2}\cdot\left(1+\sum_{n\geq 1}\frac{(-1)(-2)(-3)\dots(-1-(n-1))}{n!}\cdot(z-1)^n\right) =$

$= \frac{1}{(z-1)^2}\cdot\left(1+\sum_{n\geq 1}\frac{(-1)^n\cancel{n!}}{\cancel{n!}}(z-1)^n\right) =$

$= \frac{1}{(z-1)^2}\cdot\sum_{n\geq 0}(-1)^n(z-1)^n =$

$= \sum_{n\geq 0}(-1)^n\cdot(z-1)^{n-2}$

- **Reziduuri. Teorema reziduurilor**

1. Să se calculeze următoarele integrale complexe:

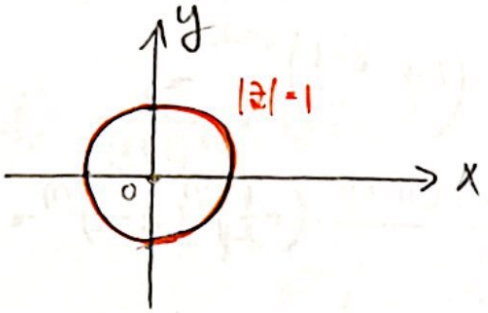
(a) $\int_{|z|=1}\frac{\operatorname{ch} z}{z}dz$ (c) $\int_{|z|=R}\frac{1}{z^2+1}dz$; $R\neq 1$ (e) $\int_{|z|=R}\frac{1}{(z+1)^2(z+3)}dz$ $R\neq 1,3$

(b) $\int_{|z|=1}\frac{\sin z}{z}dz$ (d) $\int_{|z|=R}\frac{e^{iz}}{z^2-\pi^2}dz$; $R\neq\pi$ (f) $\int_{|z|=4}\frac{1}{1+e^{z^2}}dz$.

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Rezolvare

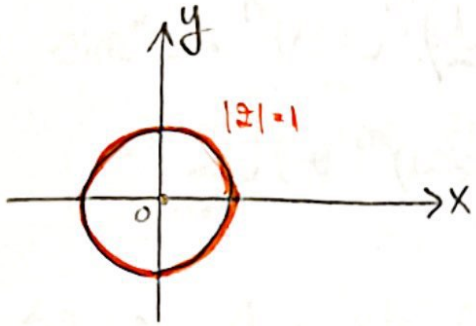
(a) $\int_{|z|=1} \frac{\operatorname{ch} z}{z} dz$; ~~$f(z) = \frac{\operatorname{ch} z}{z}$ este olomorfă~~



$\int_{|z|=1} \frac{\operatorname{ch} z}{z} dz$ ($g(z)$) Cauchy $2\pi i\, g(0) =$

$= 2\pi i \operatorname{ch}(0) = 2\pi i$

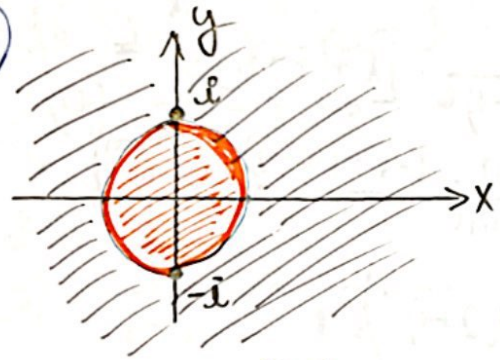
(b)



$\int_{|z|=1} \frac{\sin z}{z} dz$ ($g(z)$) Cauchy $2\pi i\, g(0) =$

$= 2\pi i \sin(0) = 0$

(c)



caz I $R < 1$

$\Rightarrow \int_{|z|=R} \frac{1}{z^2+1} dz$ Th Cauchy 0

caz II $R > 1$

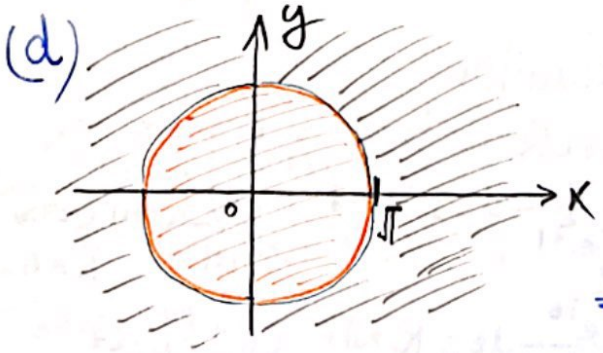
$\Rightarrow \int_{|z|=R} \frac{1}{z^2+1} dz = \int_{|z|=R} \frac{1}{(z-i)(z+i)} dz =$

$= \int_{\gamma_1} \frac{\frac{1}{z-i}}{z+i} dz + \int_{\gamma_2} \frac{\frac{1}{z+i}}{z-i} dz = 2\pi i \left(f(-i) + g(i) \right) =$

($f(z)$, $g(z)$)

$= 2\pi i \left(-\frac{1}{2i} + \frac{1}{2i} \right) = 0$

(d)



caz I $R < \pi$

$\Rightarrow \int_{|z|=R} \frac{e^{iz}}{z^2 - \pi^2} dz$ Th Cauchy 0

caz II $R > \pi$

$\Rightarrow \int_{|z|=R} \frac{e^{iz}}{z^2 - \pi^2} dz$ Th reziduurilor

$= 2\pi i \left(\operatorname{Rez}(f, \pi) + \operatorname{Rez}(f, -\pi) \right).$

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$$\text{Rez}(f,\pi) = \lim_{z\to\pi}(z-\pi)\cdot f(z) = \lim_{z\to\pi}(z-\pi)\cdot\frac{e^{iz}}{z^2-\pi^2} =$$

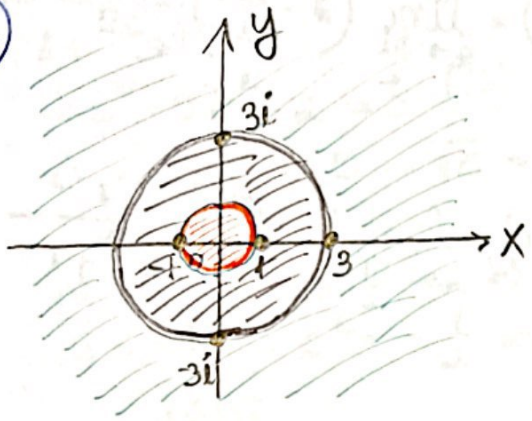
$$= \lim_{z\to\pi}\frac{e^{iz}}{z+\pi} = \frac{e^{i\pi}}{2\pi} = \frac{\cos\pi+i\sin\pi}{2\pi} = \frac{-1}{2\pi}$$

$$\text{Rez}(f,-\pi) = \lim_{z\to-\pi}(z+\pi)f(z) = \lim_{z\to-\pi}(z+\pi)\frac{e^{iz}}{z^2-\pi^2} =$$

$$= \lim_{z\to-\pi}\frac{e^{iz}}{z-\pi} = \frac{e^{-i\pi}}{-2\pi} = \frac{\cos\pi-i\sin\pi}{-2\pi} = \frac{1}{2\pi}$$

$$\Rightarrow \underline{I} = 2\pi i\cdot 0 = 0.$$

(e)



caz I: $R<1$

$$\Rightarrow \int_{|z|=R}\frac{1}{(z+1)^2(z^2+9)}dz \quad \underline{\text{Th Cauchy}} \quad 0$$

caz II $\quad 1<R<3$

$$\Rightarrow \int_{|z|=R}\frac{1}{(z+1)^2(z^2+9)}dz \quad \underline{\text{Th reziduriilor}}$$

$$= 2\pi i\,\text{Rez}(f,-1)$$

$$\text{Rez}(f,-1) = \lim_{z\to-1}\left[(z+1)^2 f(z)\right]' = \lim_{z\to-1}\left[(z+1)^2\cdot\frac{1}{(z+1)^2(z^2+9)}\right]' =$$

$$= \lim_{z\to-1}\left(\frac{1}{z^2+9}\right)' = \lim_{z\to-1}\frac{-2z}{(z^2+9)^2} = \frac{1}{50}$$

$$\Rightarrow \int_{|z|=R} f(z)dz = 2\pi i\cdot\frac{1}{50} = \frac{\pi i}{25}$$

caz III $\quad R>3$

$$\Rightarrow \int_{|z|=R} f(z)dz = 2\pi i\left[\text{Rez}(f,-1)+\text{Rez}(f,3i)+\text{Rez}(f,-3i)\right]$$

$$\text{Rez}(f,3i) = \lim_{z\to3i}(z-3i)f(z) = \lim_{z\to3i}(z-3i)\frac{1}{(z+1)^2(z^2+9)} =$$

$$= \lim_{z\to3i}\frac{1}{(z+1)^2(z+3i)} = \frac{-1}{36+48i}$$

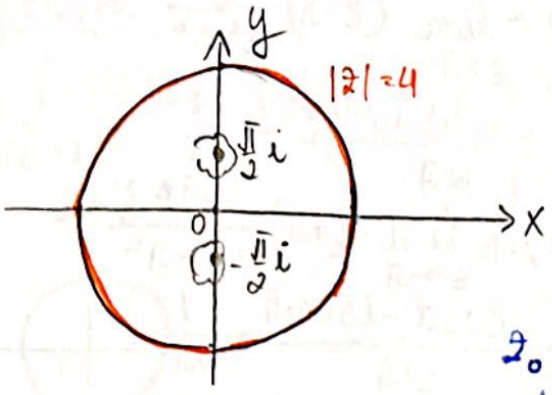
$$\text{Rez}(f,-3i) = \lim_{z\to-3i}(z+3i)f(z) = \lim_{z\to-3i}(z+3i)\frac{1}{(z+1)^2(z^2+9)} =$$

$$= \lim_{z\to3i}\frac{1}{(z+1)^2(z-3i)} = \frac{1}{36+48i}$$

$$\Rightarrow \underline{I} = 2\pi i\left(\frac{1}{50}+\frac{1}{36+48i}-\frac{1}{36+48i}\right) = \frac{\pi i}{25}.$$

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(f)



$e^{2z} = -1 \Rightarrow 2z = \operatorname{Ln}(-1) \Rightarrow$

$\Rightarrow z = \frac{1}{2}\operatorname{Ln}(-1) \Rightarrow$

$\Rightarrow z = \left\{ \frac{1}{2}\ln|-1| + \frac{1}{2}i(\operatorname{Arg}(-1) + 2k\pi) \mid k \in \mathbb{Z} \right\}$

$\Rightarrow z = \left\{ \frac{1}{2}i\pi(2k+1) \mid k \in \mathbb{Z} \right\}$

$z_0 = \frac{\pi}{2}i$; $z_{-1} = -\frac{\pi}{2}i$ se găsesc în interiorul discului.

$$\Rightarrow \int_{|z|=4} f(z)\,dz = 2\pi i \left(\operatorname{Rez}(f, z_0) + \operatorname{Rez}(f, z_{-1}) \right)$$

$$\operatorname{Rez}(f, z_0) = \lim_{z \to \frac{\pi i}{2}} \left(z - \frac{\pi i}{2}\right) f(z) = \lim_{z \to \frac{\pi i}{2}} \left(z - \frac{\pi i}{2}\right) \cdot \frac{1}{e^{2z}+1} =$$

$$\overset{\frac{0}{0}}{\underset{\text{L'Hos}}{=}} \lim_{z \to \frac{\pi i}{2}} \frac{1^2}{2e^{2z}} = -\frac{1}{2}$$

$$\operatorname{Rez}(f, z_{-1}) = \lim_{z \to -\frac{\pi i}{2}} \left(z + \frac{\pi i}{2}\right) f(z) = \lim_{z \to -\frac{\pi i}{2}} \left(z + \frac{\pi i}{2}\right) \cdot \frac{1}{e^{2z}+1} =$$

$$\overset{\frac{0}{0}}{\underset{\text{L'Hos}}{=}} \lim_{z \to -\frac{\pi i}{2}} \frac{1}{2e^{2z}} = -\frac{1}{2}$$

$$\Rightarrow I = 2\pi i \left(-\frac{1}{2} - \frac{1}{2}\right) = -2\pi i$$

(2.) Utilizând teorema reziduurilor, calculați următoarele integrale reale:

(a) $\int_0^{2\pi} \frac{1}{1+8\cos^2 t}\,dt$

(b) $\int_0^{2\pi} \frac{\sin^2 t}{5+4\cos t}\,dt$

(c) $\int_0^{2\pi} \frac{1}{1+a\cos t}\,dt$

(d) $\int_0^{\infty} \frac{x^2+1}{x^4+1}\,dx$

(e) $\int_0^{\infty} \frac{x^4}{x^6+1}\,dx.$

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Rezolvare

(a) $\int_0^{2\pi} \frac{1}{1+8\cos^2 t}\,dt$

Folosim $z = e^{it}$; $|z| = 1$; $dt = \frac{dz}{iz}$; $\cos z = \frac{1}{2}\left(z + \frac{1}{z}\right)$

$$\Rightarrow I = \int_{|z|=1} \frac{1}{1+8\cdot\frac{1}{4}\left(z+\frac{1}{z}\right)^2}\cdot\frac{1}{iz}\cdot dz = \frac{1}{i}\int_{|z|=1}\frac{1}{1+2z^2+\frac{2}{z^2}+4}\cdot\frac{1}{z}\,dz =$$

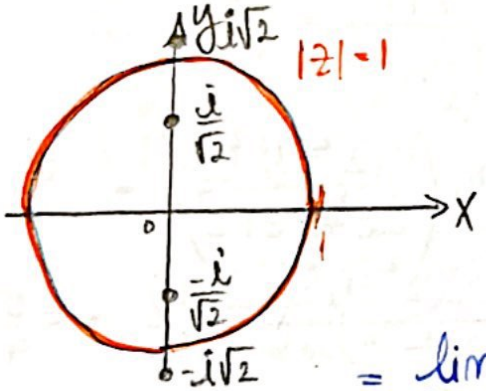
$$= \frac{1}{i}\int_{|z|=1}\frac{1}{5z+2z^3+\frac{2}{z}}\,dz = \frac{1}{i}\int_{|z|=1}\frac{z}{2z^4+5z^2+2}\,dz.$$

$2z^4+5z^2+2=0$ Notăm $z^2=u$ $\Rightarrow 2u^2+5u+2=0$

$\Delta = 25-16 = 9 \Rightarrow \sqrt{\Delta} = 3 \Rightarrow u_{1/2} = \frac{-5\pm 3}{4}$ $\rightarrow u_1 = -2$; $\downarrow u_2 = -\frac{1}{2}$

$u = -2 \Rightarrow z^2 = -2 \Rightarrow z = \sqrt{-2} = \pm i\sqrt{2}$

$u = -\frac{1}{2} \Rightarrow z^2 = -\frac{1}{2} \Rightarrow z = \sqrt{-\frac{1}{2}} = \pm\frac{i}{\sqrt{2}}$



$$\Rightarrow I = 2\pi i\left(\text{Rez}\left(f, \frac{i}{\sqrt{2}}\right) + \text{Rez}\left(f, \frac{-i}{\sqrt{2}}\right)\right)\cdot\frac{1}{i}$$

$$\text{Rez}\left(f, \frac{i}{\sqrt{2}}\right) = \lim_{z\to\frac{i}{\sqrt{2}}}\left(z-\frac{i}{\sqrt{2}}\right)f(z) =$$

$$= \lim_{z\to\frac{i}{\sqrt{2}}}\left(z-\frac{i}{\sqrt{2}}\right)\cdot\frac{z}{2z^4+5z^2+2} =$$

$$= \lim_{z\to\frac{i}{\sqrt{2}}}\frac{z}{2\left(z+\frac{i}{\sqrt{2}}\right)(z-i\sqrt{2})(z+i\sqrt{2})} =$$

$$= \frac{\frac{i}{\sqrt{2}}}{2\cdot i\sqrt{2}\cdot\frac{2i}{\sqrt{2}}\cdot\frac{3i}{\sqrt{2}}} = \frac{1}{6}$$

$$\text{Rez}\left(f, -\frac{i}{\sqrt{2}}\right) = \lim_{z\to-\frac{i}{\sqrt{2}}}\left(z+\frac{i}{\sqrt{2}}\right)f(z) = \lim_{z\to\frac{i}{\sqrt{2}}}\left(z+\frac{i}{\sqrt{2}}\right)\cdot\frac{z}{2z^4+5z^2+2} =$$

$$= \lim_{z\to-\frac{i}{\sqrt{2}}}\frac{z}{2\left(z-\frac{i}{\sqrt{2}}\right)(z-i\sqrt{2})(z+i\sqrt{2})} = \frac{-\frac{i}{\sqrt{2}}}{2\cdot i\sqrt{2}\cdot\frac{-3i}{\sqrt{2}}\cdot\frac{i}{\sqrt{2}}} = \frac{1}{6}$$

$$\Rightarrow I = \frac{4i\pi}{6} = \frac{2i\pi}{3}$$

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(c) $\int_0^{2\pi} \frac{\sin^2 t}{5+4\cos t}\,dt$.

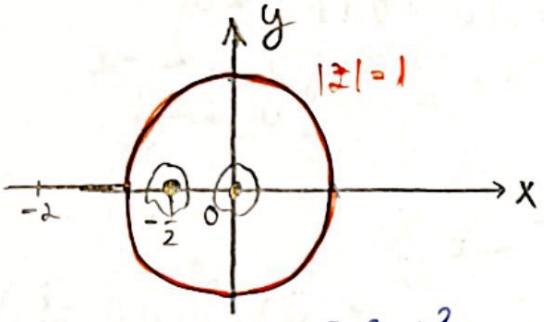
Folosim $z = e^{it}$; $|z|=1$; $dt = \frac{dz}{iz}$; $\cos t = \frac{1}{2}(z+\frac{1}{z})$; $\sin t = \frac{1}{2i}(z-\frac{1}{z})$

$$\Rightarrow I = \int_{|z|=1} \frac{\frac{-1}{4}(z-\frac{1}{z})^2}{5+4\cdot\frac{1}{2}(z+\frac{1}{z})} \cdot \frac{1}{iz}\,dz = -\frac{1}{4i}\int_{|z|=1} \frac{z^2+\frac{1}{z^2}-2}{5z+2z^2+2}\,dz =$$

$$= -\frac{1}{4i}\int_{|z|=1} \frac{(z^2-1)^2}{z^2(2z^2+5z+2)}\,dz.$$

$2z^2+5z+2=0$

$\Delta = 25-16 = 9 \Rightarrow \sqrt{\Delta} = 3 \Rightarrow z_{1,2} = \frac{-5\pm 3}{2} \Rightarrow z_1 = -2$; $z_2 = -\frac{1}{2}$



$$\Rightarrow I = 2\pi i\left(\operatorname{Rez}(f,0) + \operatorname{Rez}(f,-\tfrac{1}{2})\right)\left(-\tfrac{1}{4i}\right)$$

$$\operatorname{Rez}(f,-\tfrac{1}{2}) = \lim_{z\to -\frac{1}{2}} (z+\tfrac{1}{2})\,f(z) =$$

$$= \lim_{z\to -\frac{1}{2}} (z+\tfrac{1}{2})\cdot\frac{(z^2-1)^2}{z^2(2z^2+5z+2)} =$$

$$= \lim_{z\to -\frac{1}{2}} \frac{(z^2-1)^2}{2z^2(z+2)} = \frac{3}{4}$$

$$\operatorname{Rez}(f,0) = \lim_{z\to 0}\left[z^2 f(z)\right]^{(1)} = \lim_{z\to 0}\left(\frac{(z^2-1)^2}{2z^2+5z+2}\right)' =$$

$$= \lim_{z\to 0} \frac{2(z^2-1)\cdot 2z(2z^2+5z+2) - (z^2-1)^2(4z+5)}{(2z^2+5z+2)^2} = -\frac{5}{4}$$

$$\Rightarrow I = 2\pi i\cdot\frac{-1}{4i}\left(\frac{3}{4}-\frac{5}{4}\right) = \frac{-\pi}{2}\cdot\frac{-1}{2} = \frac{\pi}{4}$$

(d) $\int_0^{\infty} \frac{x^2+1}{x^4+1}\,dx = \frac{1}{2}\int_{-\infty}^{\infty} \frac{x^2+1}{x^4+1}\,dx = \frac{1}{2}\sum_{\operatorname{Im}(z_k)>0} \left(\operatorname{Rez} f, z_k\right)\cdot 2\pi i$

$f(z) = \frac{z^2+1}{z^4+1}$; $z^4 = -1 = e^{i(2k+1)\pi} \Rightarrow z_k = e^{\frac{i(2k+1)\pi}{4}}$; $k\in\{0,1,2,3\}$

$z_0 = e^{i\pi/4}$; $z_1 = e^{i3\pi/4}$; $z_2 = e^{i5\pi/4}$; $z_3 = e^{i7\pi/4}$

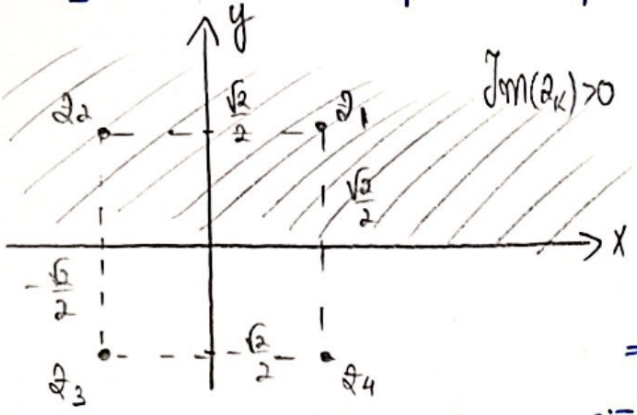
$z_0 = e^{i\pi/4} = \cos\frac{\pi}{4} + i\sin\frac{\pi}{4} \Rightarrow \operatorname{Im}(z_0)>0$

$z_1 = e^{3i\pi/4} = \cos\frac{3\pi}{4} + i\sin\frac{3\pi}{4} \Rightarrow \operatorname{Im}(z_1)>0$

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$z_2 = e^{5\pi i/4} = \cos\frac{5\pi}{4} + i\sin\frac{5\pi}{4} \Rightarrow \operatorname{Im}(z_2) < 0$

$z_3 = e^{7\pi i/4} = \cos\frac{7\pi}{4} + i\sin\frac{7\pi}{4} \Rightarrow \operatorname{Im}(z_3) < 0$



$$\Rightarrow \int_{-\infty}^{\infty} \frac{x^2+1}{x^4+1}\,dx = \left[\operatorname{Rez}(f, z_0) + \operatorname{Rez}(f, z_1)\right] \cdot 2\pi i$$

$$\operatorname{Rez}(f, z_k) = \frac{z_k^2+1}{4z_k^3}$$

$$\Rightarrow \operatorname{Rez}(f, z_0) = \frac{e^{i\pi/2}+1}{4e^{3i\pi/4}}$$

$e^{i\pi/2} = \cos\frac{\pi}{2} + i\sin\frac{\pi}{2} = i$; $e^{3i\pi/4} = \cos\frac{3\pi}{4} + i\sin\frac{3\pi}{4} = -\frac{\sqrt{2}}{2} + i\frac{\sqrt{2}}{2}$

$$\Rightarrow \operatorname{Rez}(f, z_0) = \frac{i+1}{4\left(-\frac{\sqrt{2}}{2}+i\frac{\sqrt{2}}{2}\right)} = \frac{i+1}{2\sqrt{2}(i-1)}$$

$$\operatorname{Rez}(f, z_1) = \frac{e^{3i\pi/2}+1}{4 \cdot e^{9i\pi/4}}$$

$e^{3i\pi/2} = \cos\frac{3\pi}{2} + i\sin\frac{3\pi}{2} = -i$; $e^{9i\pi/4} = \cos\frac{9\pi}{4} + i\sin\frac{9\pi}{4} = \frac{\sqrt{2}}{2} + i\frac{\sqrt{2}}{2}$

$$\Rightarrow \operatorname{Rez}(f, z_1) = \frac{-i+1}{4\left(\frac{\sqrt{2}}{2}+i\frac{\sqrt{2}}{2}\right)} = \frac{1-i}{2\sqrt{2}(i+1)}$$

$$\Rightarrow \int_{-\infty}^{\infty} \frac{x^2+1}{x^4+1}\,dx = \frac{i+1}{2\sqrt{2}(i-1)} + \frac{1-i}{2\sqrt{2}(i+1)} = \frac{(i+1)^2+(1-i)(i-1)}{2\sqrt{2}(i-1)(i+1)} =$$

$$= \frac{-1+2i+1+i+1-i}{2\sqrt{2}(i^2-1)} = \frac{4i}{-4\sqrt{2}} = \frac{-i}{\sqrt{2}}$$!?! (trebuia fără i)

$$\Rightarrow \int_0^{\infty} \frac{x^2+1}{x^4+1}\,dx = \frac{1}{2} \cdot 2\pi i \cdot \frac{-i}{\sqrt{2}} = \frac{\pi}{\sqrt{2}}.$$

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